

Giuliana Cabrera Sanchez

Boston, MA • gpcs@mit.edu • [linkedin.com/in/giuliana-cabrera/](https://www.linkedin.com/in/giuliana-cabrera/) • [gcabreraa.github.io](https://github.com/gcabreraa)

EDUCATION

Massachusetts Institute of Technology (MIT) | Cambridge, MA

Candidate for B.S. in Computer Science and Electrical Engineering

Class of May 2025

Relevant Courses: Digital Systems, Microcomputer Project Laboratory (Circuit Design, Chips, Microcontrollers),

Computation Structures (RISC-V, Minispec), Software Construction, Algorithms, Digital Systems (FPGAs)

EXPERIENCE

Research Intern, Queen's University Belfast | Belfast, UK

Sept 2025 - Dec 2025

- Integrated URJA, a shared-library tool for dynamic voltage and frequency scaling performance counter collection, as a SPANK plug-in for the Slurm job manager, enabling node-level energy profiling and runtime frequency control.
- Developed shell scripts and automation workflows on Linux to streamline deployment, logging, and system testing across compute nodes.

Software Development Intern, Qualcomm | San Diego, CA

June 2025 - Aug 2025

- Designed a Jupyter-based profiling framework for Snapdragon SoCs, enabling visualization of power draw, battery drain, and cache-stall metrics collected via on-chip performance counters.
- Automated data ingestion and analysis pipelines to evaluate CPU/GPU workload traces, identifying opportunities to refine dynamic clock and voltage scaling algorithms used.

Undergraduate Researcher, MIT Global Engineering and Research Lab | Cambridge, MA

Feb 2025 - May 2025

- Designed and programmed hardware drivers in Python for sensors and actuators within an off-grid, containerized electrodialysis desalination system.
- Implemented and debugged communication protocols, including RS-485 and MQTT, to enable a reliable supervisory control of the pumps, valves, and sensors system with automated testing scripts.

Software Engineering Intern, Qualcomm | San Diego, CA

June 2024 - Aug 2024

- Developed a new raytracing feature through bounding volume hierarchies as part of the graphics tools team for Qualcomm's Snapdragon Profiler.

Software Engineering Intern, Slack (Salesforce) | San Francisco, CA

June 2023 - Aug 2023

- Enhanced Slack's calling function by introducing a native menu feature for macOS in Typescript and Redux, building it on the Electron framework.

Hardware Development Intern, IBM Accelerate Program | Remote

June 2022 - Aug 2022

- Authored a collaborative group paper that delved into the realm of general-purpose and embedded systems.

PROJECTS

MITOSIS Sensor Node (6.900 Final Project – Firmware Lead)

May 2025

- Led firmware design for a compact solar-powered temperature and humidity sensing device using ESP32-C3, e-ink display, and LoRa mesh networking.
- Implemented fault/error detection, deep sleep scheduling, and RTC-based data archiving (~15-day buffer).
- Integrated sensor drivers (SHT40, VEML7700) and display refresh logic in C++, optimizing for low power and modular expansion.

PSoC 5LP Piano Tiles

May 2024

- Developed an interactive Piano Tiles game using a PSoC microcontroller, integrating a TFT display for gameplay.
- Programmed game logic in C and Assembly for smooth display scrolling, accurate button response, including debouncing techniques, and synthesized audio, using an 8051 microcontroller, DAC, EEPROM, and LM386 amplifier..

Smart Dungeons and Dragons Game

May 2022

- Engineered essential player functionalities, including dice rolling, combat actions like stabbing, and defensive maneuvers such as shield usage, all coordinated through ESP32 devices, for an interactive Dungeons and Dragons game.

LEADERSHIP / VOLUNTEERING

Vice President, MIT Latino Cultural Center

Sept 2022 - May 2025

Secretary, MIT Society of Hispanic Professional Engineers

Aug 2021 - May 2025

Former Vice President of DEI and President, MIT Alpha Chi Omega

Dec 2022 - Dec 2024

SKILLS

Programming Languages: Python, C/C++, Typescript, SystemVerilog, Minispec, Bash, Assembly Language

Other: Git, TeraTerm, ExpressPCB, kiCAD, Ubuntu/Linux, Docker, SSH, UART, MQTT

HONORS Member of Rewriting the Code + ColorStack, The Jack Kent Cooke Foundation College Scholar, Hispanic Scholarship Fund Scholar